GenAI/AGENTS

Applied Agentic AI

Introduction

Instructor Profile

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Audience Profile

What is your expectation from this course ?

Types of GenAI Jobs

Startup/Agency

Jobs would involve part time or fulltime creation of videos, copy ads, blog post, creative banners etc, using tools and SaaS platforms.
Good prompting skill is sufficient, pay will be less with low job security.

Product/Enterprise MNC

Jobs would involve fulltime working as GenAI/Agentic Engineer trying to find new use cases for increasing revenue, increasing productivity, faster mordernization of existing legacy systems, complex integrations with other enterprise solutions into existing IT landscape.
Good practical knowledge needed, high pay, high growth, diverse skillset ranging from GenAI to Data Science to MLOps (overlapping skills and job role)

Demo/Use Cases

1. **Chat System with Agentic RAG**
2. **UI/UX Prototyping Agent**
3. **Code to Document Generation Agent**
4. **Code Review Agent**
5. **Specifiction to Test Cases Generation Agent**
6. **Technical Analysis Stock Market Agent**
7. **Github CoPilot Agents**

Course Curricullum

- Evolution of Artificial Intelligence
- Supervised/Unsupervised Learning
- Deep Learning
- Transformers & GPT
- GenAI, LLM and different models & types(Language, Vision, Audio)
- Prompt Enginneering = System vs User
- Practical Python
- RAG systems
- Agents and different Agentic frameworks
- Development & Deployment of Dockerized Multi Agentic System
- MCP Protocol, developing & deploying MCP servers
- Huggingface, Openrouter, Groq cloud etc
- Real world Enterprise Use Case to implement in any project

WHY LEARN GenAI now ?

- Disruption is real
- Powerful automation plugins and tools getting announced everyday
- Companies contracts with clients are changing, moving away from hourly model of business to output based delivery.
- Billions of $ getting invested for expansions.
- According to World Economic Forum has projected, 92 millions jobs will disappear but 172 million jobs will be created
- Traditional IT services & BPO jobs will disappear fast.
- Companies are already planning for mass layoffs to compensate their losses during current/future quaterly results.

Company’s expectation from You?

- Management roles must learn to lead teams who will develop solutions using AI, that could help provide value to client.
- No more just writting prompts and getting answers and copy pasting, instead create workflows and automations pipeline.
- As junior to mediore no more just coding, instead you must also learn Architecture or Cloud or AI along with coding.
- Seniors must make sure the overall productivity and quality of team increases with few employees.
- Company have aleardy rolled out GHCP/Claude inside their network and expects employees to use them and deliver efficiently.

Career Options & Expected CTC?

- Data Sceince
- ML Engineer
- MLOps Engineer
- GenAI/Agentic Developer
- AI Architect
- Your own Startup
- Average CTC could be around 35 LPA for 5+ years of experience

How to GET/SWITCH to AI Jobs?

- Crystal clear understanding of GenAI & Agents
- Building Portfolio with real world projects
- Solving day to day problems at your current job with GenAI
- Develop Agents for getting new businesses/revenue for your company
- Develop Agents for various SDLC life cycle stages
- Develop Agents for OPS, Infra and Release process along with Incident management.
- Participating in hackathons, events, writing linkedin posts
- Doing certifications for Azure Cloud around AI/GenAI/Agents
- Implementing MLOps for Agents/RAG/Fine tuning systems

Pillars of Intelligence

Data

Learning

Reasoning

Taking Action

What is AI

AI, or Artificial Intelligence, is when computers or machines are made to think and act like humans. This means they can learn from experience, understand language, make decisions, and solve problems—just like people do.

For example:

- When your phone understands your voice and answers questions, that’s AI.
- When Netflix recommends a show you might like, it’s using AI to guess your preferences.

It’s like giving machines a kind of "smart brain" to help them do tasks without needing to be told every little step.

What is Machine Learning

Machine Learning (ML) is a part of Artificial Intelligence where computers learn from data—just like humans learn from experience—without being directly programmed for every task.

In simple words:

- You give the computer lots of examples (data).
- It looks for patterns in that data.
- Then it uses those patterns to make predictions or decisions on new data.

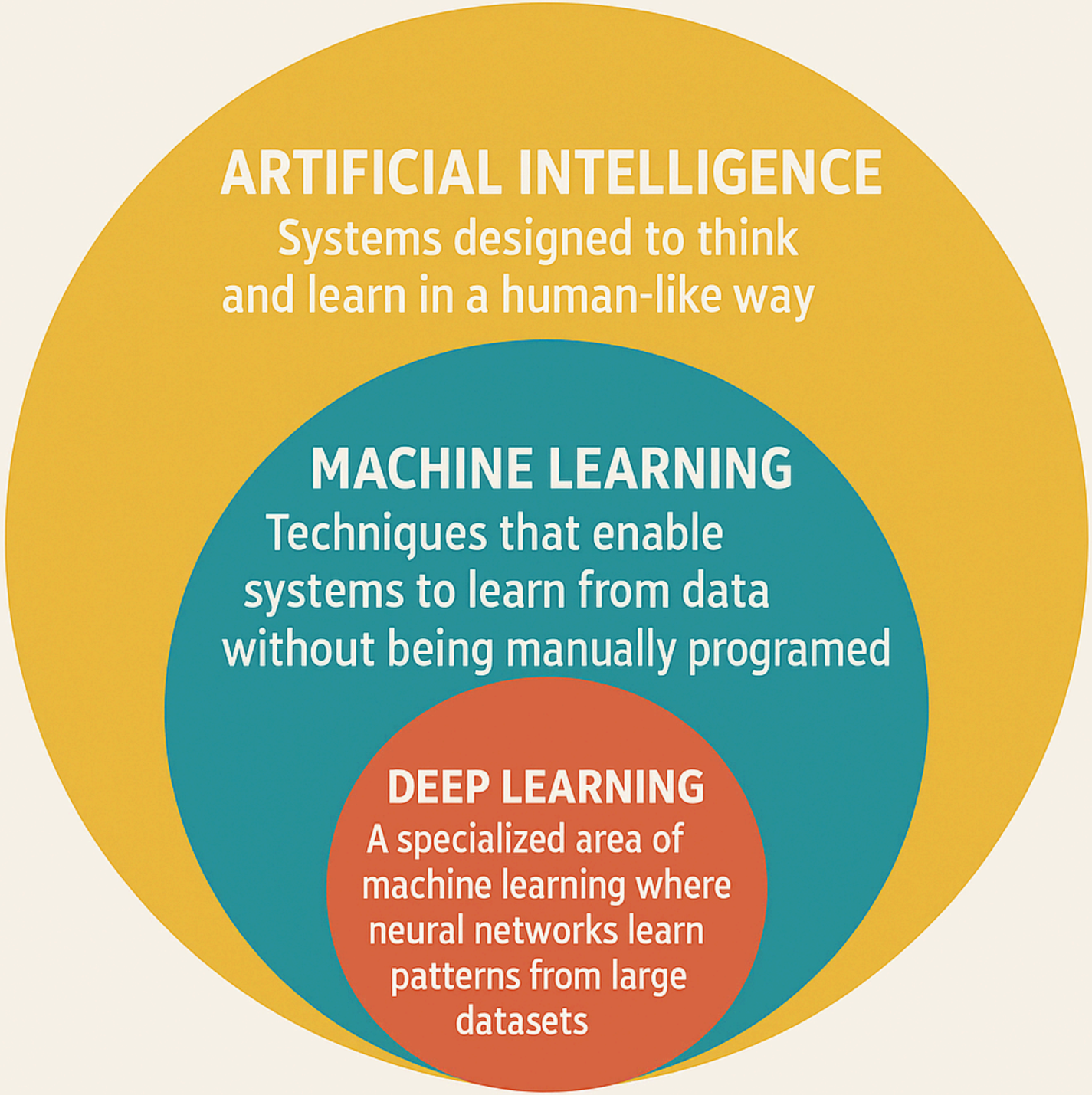
Example:

If you want a computer to tell whether a photo is of a cat or a dog:

- You show it thousands of labeled photos of cats and dogs.
- It learns the differences between them.
- Later, you give it a new photo, and it can guess if it’s a cat or a dog.

It’s like teaching a child by showing examples instead of giving strict instructions.

Relationship



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Artificial Intelligence
│
└── Machine Learning
    │
    ├── Supervised Learning
    ├── Unsupervised Learning
    ├── Reinforcement Learning
    │
    └── Deep Learning
        │
        └── Transformer Architecture
            │
            └── GPT Models
```

Supervised Learning – Learning With a Teacher

A type of machine learning where the model learns from labeled data.
👉 Input + Correct Output are provided during training.

Ravi's teacher shows him:
🍎 "This is an Apple."
🐶 "This is a Dog."
Every time Ravi answers wrong, the teacher corrects him.
👉 Ravi learns because correct answers are given.
💡 AI Meaning:
Data + Correct Answer (Label)
Model learns from examples.
📌 Like:
Email → Spam or Not Spam

Unsupervised Learning – Learning Without a Teacher

Model learns from unlabeled data.

👉 No correct answers given.

📌 Example:

- Customer segmentation
- Market basket analysis
- Pattern discovery

Ravi gets a box of mixed toys:

🚗 Cars, 🧸 Teddy bears, 🏀 Balls.

No one tells him what they are.

He groups:

All cars together

All soft toys together

All balls together

👉 No teacher. He finds patterns himself

.

💡 AI Meaning:

- No labels.
- Model finds hidden groups (clusters).

Deep Learning – Big Brain with Many Layers

When Ravi sees a dog, his brain thinks:

- Shape?
- Color?
- Sound?
- Tail?
- Movement?

Many thinking layers happen inside his brain.

👉 The more he practices, the smarter he gets.

💡 AI Meaning:

Deep learning uses many neural network layers to learn complex things like:

- Faces
- Voice
- Language

Deep Learning

A subset of machine learning using multi-layer neural networks.

👉 Inspired by the human brain.

📌 Why “Deep”?

Because it has many hidden layers.

📊 Types:

CNN → Images

RNN → Sequences

LSTM → Text

Transformers → Modern NLP

🎯 Used For:

- Speech recognition
- Self-driving cars
- Chatbots
- Face recognition

Reinforcement Learning – Learning by Rewards

Ravi is learning to ride a bicycle.

Falls ❌ → No reward

Balances properly ✅ → Gets chocolate 🍫

He learns by:

👉 Trying

👉 Failing

👉 Getting rewards

💡 AI Meaning:

AI agent learns by:

Action

Reward

Punishment

Transformer Architecture – Paying Attention

Teacher says:
“The monkey didn’t eat the banana because it was rotten.”
Ravi understands:
👉 “it” means banana (not monkey).
How?
Because he pays attention to important words.
💡 AI Meaning:
Transformer uses Self-Attention
It looks at all words and understands relationships.
It was introduced in the famous paper:
“Attention Is All You Need”

GPT Models – The Smart Storytelling Kid

Ravi reads 1000 storybooks 📚.
Now if you say:
"Once upon a time..."
He continues:
"There was a brave dragon..."
He predicts the next words!

💡 AI Meaning:
GPT = Generative Pre-trained Transformer
Built by:
OpenAI
Example:
ChatGPT

It:
Reads massive text
Learns patterns
Predicts next word
Generates stories, answers, code

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AI = Smart Robot World

Machine Learning = Robot goes to school

    Supervised = Teacher teaches robot
    Unsupervised = Robot learns alone
    Reinforcement = Robot learns by rewards

Deep Learning = Robot has big layered brain

Transformer = Robot pays attention to words

GPT = Robot becomes expert storyteller
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Supervised → Teacher teaches with answers
Unsupervised → Kid finds patterns alone
Deep Learning → Big brain with many layers
Reinforcement → Learn by reward & punishment
Transformer → Pays attention to relationships
GPT → Smart storyteller using transformer

Agentic AI – The Independent Smart Kid 🤖

Now Ravi grows up a little.

Earlier:

He only answered questions.

Now:

He starts doing tasks on his own.

Teacher says:

“Ravi, prepare a science project on solar system.”

Now Ravi:

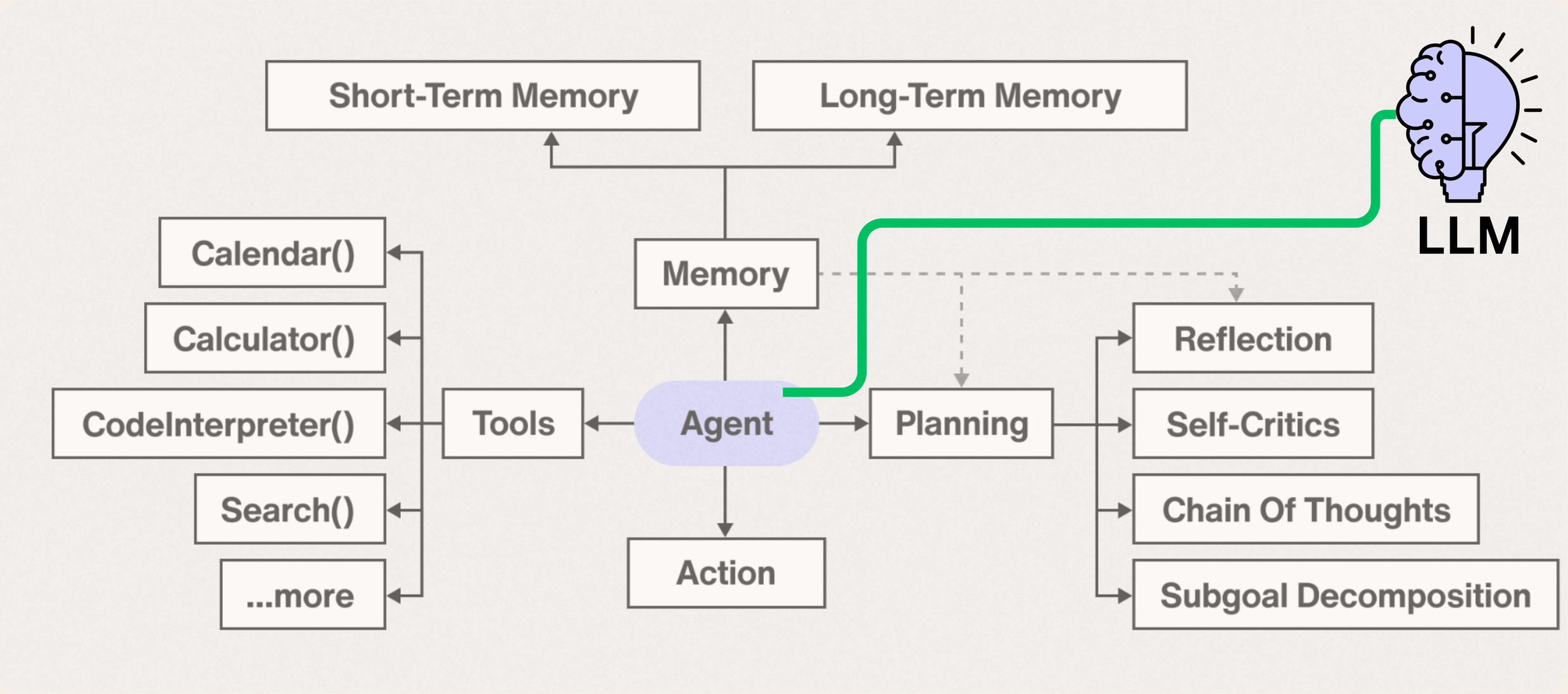
- Plans what to do
- Searches books
- Makes notes
- Draws diagrams
- Builds model
- Submits project

Teacher didn’t guide every step.

He acted independently.

That is Agentic behavior.

Parts of an AI Agent



Developing GHCP Agents

- Install Python
- Install Git
- Install VSCode
- Install Copilot VSCode Extension
- Install Markdown Viewer Enhanced VSCode extension